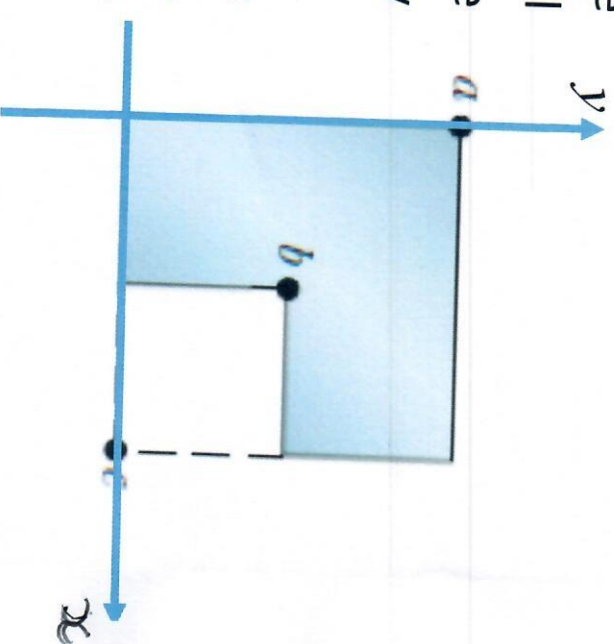


1. The figure shows a uniform metal plate that had been square before 25% of it was cut out. Find position of the centre of mass of the metal plate in the figure if the side lengths are a . Use the co-ordinate system shown and give clear reasons for any assumptions you may make.

Die figuur dui 'n uniforme metaal plaat wat oorspronklik vierkantig was voordat 25% daarvan afgesny was. Vind die posisie van die massamiddelpunt van die metaal plaat in die figuur indien die plaat se sy-lengtes het van a . Gebruik die koördinaatstelsel aangedui en gee duidelike verklarings vir enige aannames wat jy maak.

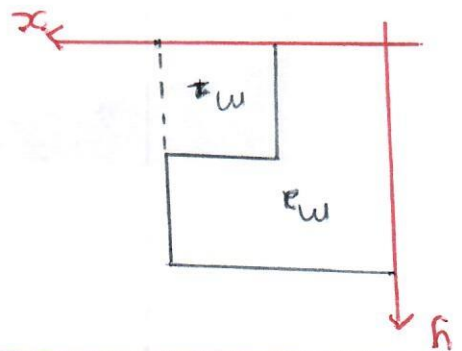


(7)

2. A 75 kg man rides on a 39 kg cart moving at a velocity of 2.3 m/s. He jumps off with zero horizontal velocity relative to the ground. What is the resulting change in the cart's velocity, including sign?

’n 75 kg man ry op ’n 39 kg wa wat teen ’n snelheid van 2.3 m/s beweeg. Jy spring van die wa af met geen horisontale snelheid ten opsigte van die grond. Wat is die verandering in die wa se snelheid, insluitend die teken?

(3)



The original square plate had a com at the centre $(\frac{a}{2}, \frac{a}{2})$ coordinates. ✓
 The cut out has a com at $(\frac{3}{4}a, \frac{1}{4}a)$ ✓
 Mass of plate initially: M
 Mass of cut out piece: $\frac{M}{4}$ (m_1)
 Mass of final piece: $\frac{3M}{4}$ (m_2)

COM of M will be the combined com of $\frac{M}{4}$ and $\frac{3M}{4}$

$$\therefore x_{com, M} = \frac{1}{M} (x_{m_1} m_1 + x_{m_2} m_2)$$

$$\frac{2}{9} = \frac{1}{M} \left(\frac{3a}{4} \cdot \frac{M}{4} + x_{m_2} \frac{3}{4} M \right) \quad \checkmark$$

$$\therefore x_{m_2} = M \left(\frac{2}{9} - \frac{3a}{4} \right) \frac{4}{3} = \frac{M}{3} \times \frac{5}{16} = \frac{5}{12} a$$

$$y_{com, M} = \frac{1}{M} (y_{m_1} m_1 + y_{m_2} m_2)$$

$$\frac{2}{9} = \frac{1}{M} \left(\frac{9}{4} \cdot \frac{M}{4} + y_{m_2} \frac{3}{4} M \right) \quad \checkmark$$

$$y_{m_2} = M \left(\frac{2}{9} - \frac{9}{4} \right) \frac{4}{3} = \frac{16}{7} \cdot \frac{1}{4} a = \frac{28}{48} a$$

$$\underline{\underline{= \frac{12}{7} a = 0.58a}} \quad \checkmark$$

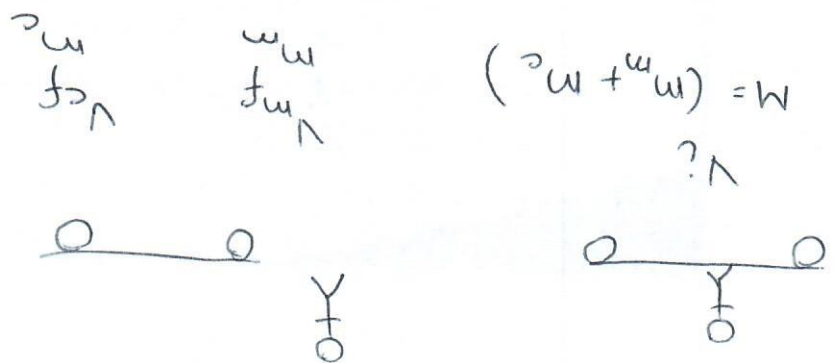
Answers can be fractions or

decimals

$\therefore$ COM coordinates: $(0.42a, 0.58a)$

Alternative method: Use 3 squares $\begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix}$ or 2 pieces $\begin{bmatrix} m_1 \\ m_2 \end{bmatrix}$

2.



$$\Delta p = 0$$

$$\therefore p_i = p_f$$

$$(m_1 + m_2) v_i = m_1 v_{1f} + m_2 v_{2f}$$

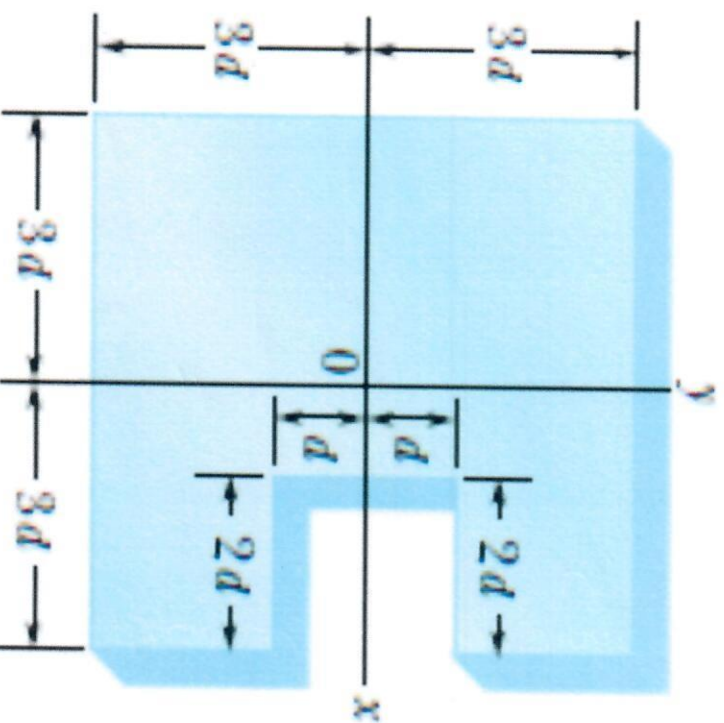
$$v_{cf} = \frac{m_2}{(m_1 + m_2)} v_i$$

$$= \frac{(75 \text{ kg} + 39 \text{ kg}) \cdot 2.3 \text{ m/s}}{39 \text{ kg}}$$

$$= 6.72 \text{ m/s} = \underline{\underline{6.7 \text{ m/s}}}$$

Change in the velocity: $\Delta v = v_f - v_i$

$$= 6.7 \text{ m/s} - 2.3 \text{ m/s} = +4.4 \text{ m/s}$$

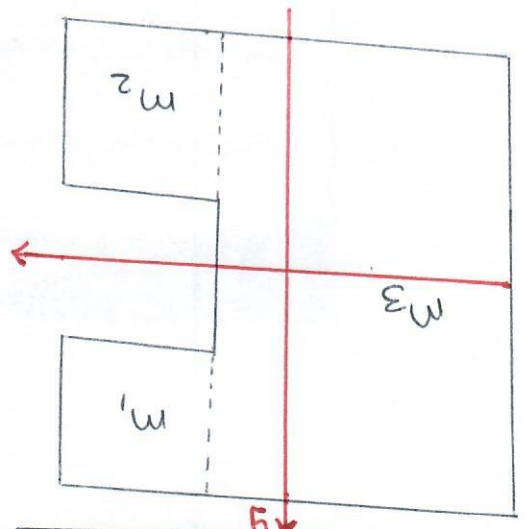


1. The figure shows a uniform square plate of edge length $6d = 6.0$ m from which a square piece of edge length $2d$ has been removed. What are the x- and y-coordinates of the centre of mass of the remaining piece?

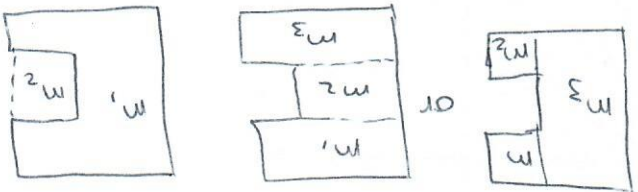
Die figuur dui 'n uniforme vierkantige plaat met kant lengte $6d = 6$ m, waarvan 'n vierkantige stuk met sy lengte $2d$ verwyder is. Wat is die x- en y-koördinate van die massa-middelpunt van die oorblywende stuk? (7)

2. A body of mass 2.0 kg makes an elastic collision with another body at rest and continues to move in the original direction but with one-fourth of its original speed. (a) What is the mass of the other body? (3)

①



1. Divide plate into different convenient parts.



2. Determine masses of each section as a fraction of the whole

Let each $2d \times 2d$ piece have a mass m
 Then $m_1 = m$, $m_2 = m$, $m_3 = 6m$

com of $m_1 = (2d, 2d)$

$m_2 = (2d, -2d)$

$m_3 = (-1d, 0)$

$$\therefore x_{com} = \frac{1}{M} (m_1 x_1 + m_2 x_2 + m_3 x_3)$$

$$= \frac{1}{8m} (m(2d) + m(2d) + 6m(-1d))$$

$$= \frac{1}{8m} (3dm) = \frac{3}{8}d$$

$$y_{com} = \frac{1}{M} (m_1 y_1 + m_2 y_2 + m_3 y_3)$$

$$= \frac{1}{8m} (m(2d) + m(-2d) + 6m(0))$$

$$= 0$$

(2)

$$\begin{aligned} & \text{○} \quad m_1 = 2.0 \text{ kg} \quad v_1 = 4 \text{ m/s} \quad v_{f1} = \frac{1}{4} v_1 \\ & \text{○} \quad m_2 \quad v_2 = 0 \quad v_{f2} = ? \end{aligned}$$

Elastic collision

can use $\Delta p = 0$ & $\Delta K = 0$

$$\therefore v_{f1} = \frac{m_1 - m_2}{m_1 + m_2} v_{i1}$$

$$v_{f1} (m_1 + m_2) = v_{i1} (m_1 - m_2)$$

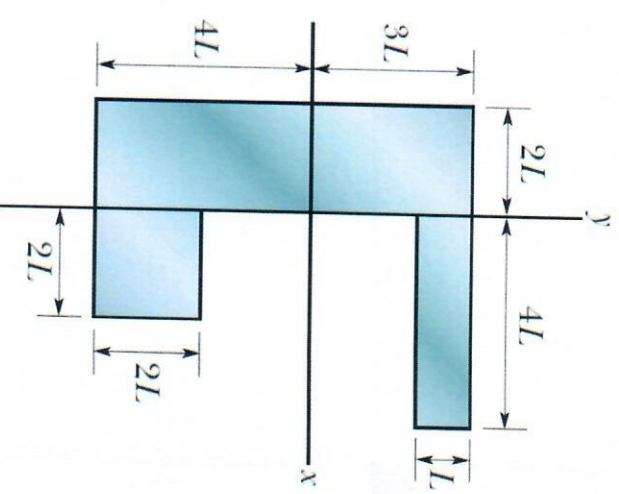
$$v_{f1} = \frac{1}{4} v_{i1}, \quad m_1 = 2 \text{ kg}$$

$$\text{solving: } m_2 = \frac{5}{3} m_1 = 1.2 \text{ kg}$$

1. What are the coordinates of the centre of mass of the uniform plate shown in the figure if $L = 5.0 \text{ cm}$?

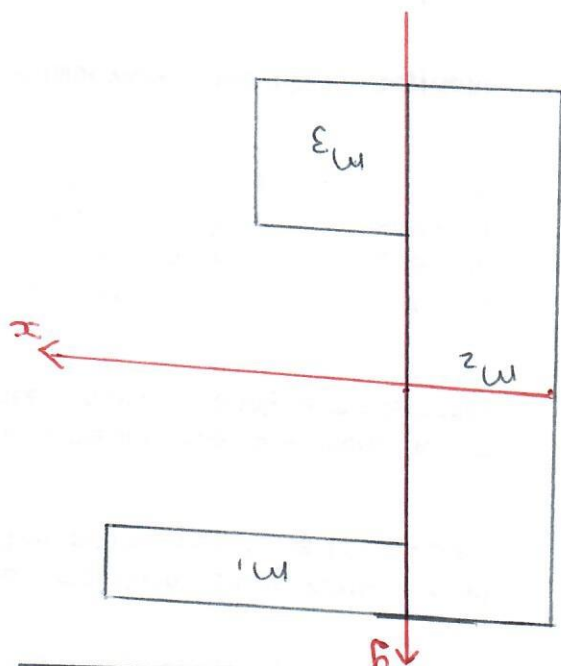
Wat is die koördinate van die massa middelpunt van die univorme plaat in die figuur, indien $L = 5.0 \text{ cm}$?

(5)



2. A 0.30 kg softball has a velocity of 15 m/s at an angle of 35° below the horizontal just before making contact with the bat. What is the magnitude and direction of the change in momentum of the ball while in contact with the bat if the ball leaves with a velocity of 20 m/s , vertically downward.
- ’n 0.3 kg bal het ’n snelheid van 15 m/s gerig teen ’n hoek van 35° onder die horisontaal net voor dit kontak maak met die kolf. Wat is die grootte en rigting van die verandering in die momentum van die bal terwyl dit in kontak is met die kolf, indien die bal dit verlaat met ’n snelheid van 20 m/s , gerig vertikaal afwaarts.*

(5)



Masses:
 $m_1 = 4m$
 $m_2 = 14m$
 $m_3 = 4m$

coms of the 3 sections:
 $com\ m_1 = (2L, 2\frac{1}{2}L)$
 $m_2 = (-1L, -\frac{2}{3}L)$
 $m_3 = (1L, -3L)$

$$x_{com} = \frac{1}{M} (m_1 x_1 + m_2 x_2 + m_3 x_3)$$

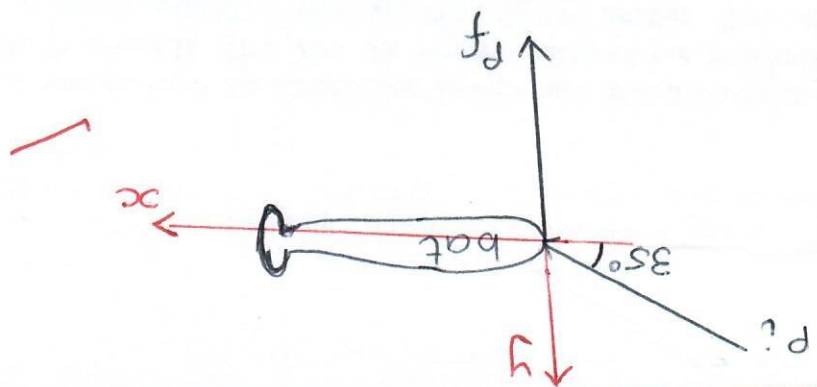
$$= \frac{1}{22m} (4m(2L) + 14m(-1L) + 4m(1L))$$

$$= \frac{8L - 14L + 4L}{22} = -\frac{2L}{22} = -0.045L \text{ (or } -\frac{10}{22}L \text{ cm)}$$

$$y_{com} = \frac{1}{M} (m_1 y_1 + m_2 y_2 + m_3 y_3)$$

$$= \frac{1}{22m} (4m(2.5L) + 14m(-\frac{2}{3}L) + 4m(-3L))$$

$$= \frac{10L - 7L - 12L}{22} = -\frac{9L}{22} = -2.0 \text{ cm (or } -\frac{45}{22}L \text{ cm)}$$



$$\Delta p_x = m v_{fx} - m v_{ix}$$

$$= m (0 - v_i \cos 35^\circ)$$

$$= 0.30 \text{ kg} (0 - 15 \text{ m/s} \cos 35^\circ)$$

$$= -3.69 \text{ kg} \cdot \text{m/s}$$

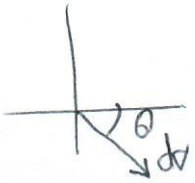
$$\Delta p_y = m v_{fy} - m v_{iy}$$

$$= m (20 \text{ m/s} - 15 \text{ m/s} \sin 35^\circ)$$

$$= 11.40 \text{ kg} \cdot \text{m/s}$$

$$|\Delta p| = \sqrt{\Delta p_x^2 + \Delta p_y^2} = 12.08 \text{ kg} \cdot \text{m/s}$$

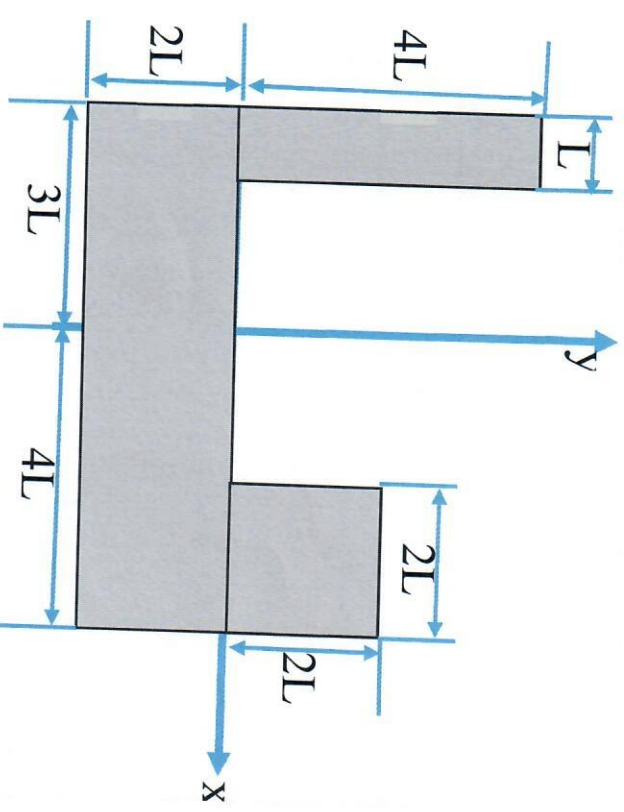
$$\theta = \tan^{-1} \left(\frac{\Delta p_y}{\Delta p_x} \right) = 72.1^\circ$$



1. What are the coordinates of the centre of mass of the uniform plate shown in the figure if $L = 5.0 \text{ cm}$?

Wat is die koördinate van die massa middelpunt van die uniforme plaat in die figuur, indien $L = 5.0 \text{ cm}$?

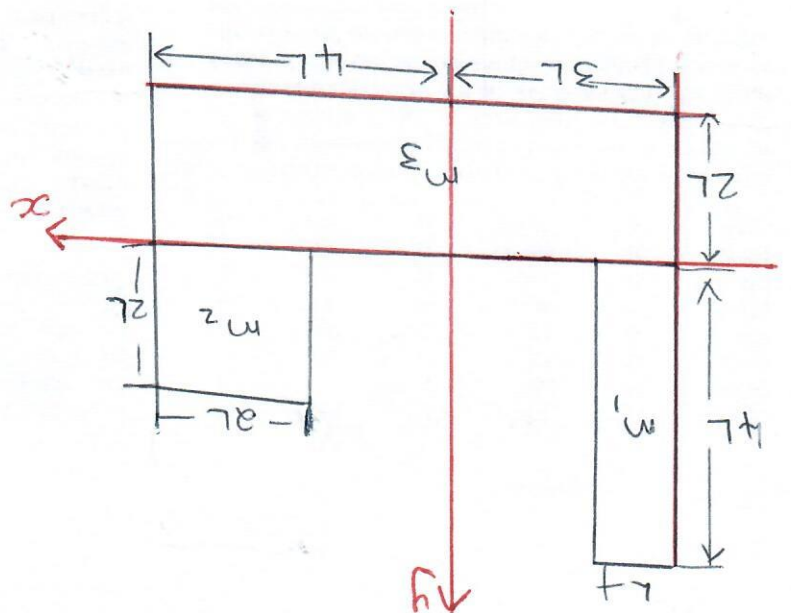
(5)



2. A 0.30 kg softball has a velocity of 15 m/s at an angle of 35° below the horizontal just before making contact with the bat. What is the magnitude and direction of the change in momentum of the ball while in contact with the bat if the ball leaves with a velocity of 20 m/s , vertically downward.

’n 0.3 kg bal het ’n snelheid van 15 m/s gerig teen ’n hoek van 35° onder die horisontaal net voor dit kontak maak met die kolf. Wat is die grootte en rigting van die verandering in die momentum van die bal terwyl dit in kontak is met die kolf, indien die bal dit verlaat met ’n snelheid van 20 m/s , gerig vertikaal afwaarts.

(5)



Masses:

$m_1 = 2m$ ✓
 $m_2 = 2m$ ✓
 $m_3 = 7m$ ✓
 or any other equivalent ratio.

coms of mass sections

$m_1 - (-2L, 2L)$ ✓
 $m_2 - (2L, 1L)$ ✓
 $m_3 - (\frac{1}{2}L, -1L)$ ✓

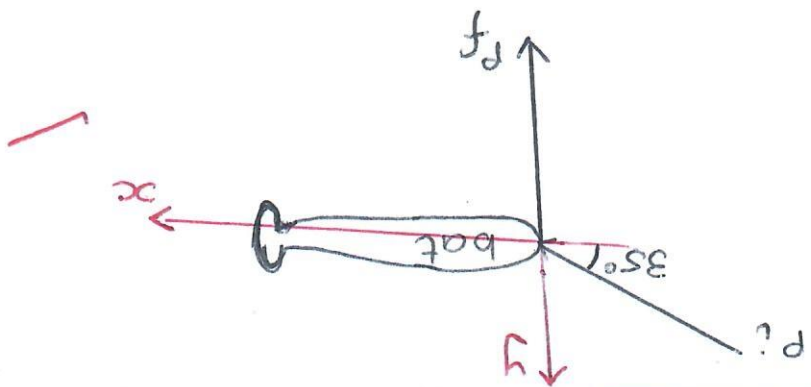
$$x_{com} = \frac{1}{M} (m_1 x_1 + m_2 x_2 + m_3 x_3) = \frac{11m}{T} (2m(-2\frac{1}{2}L) + 2m(2L) + 7m(\frac{1}{2}L))$$

$$= 4.5L = 2.0 \text{ cm} \quad \left(\text{or } 22.5 \text{ cm} \right)$$

$$y_{com} = \frac{M}{T} (m_1 y_1 + m_2 y_2 + m_3 y_3)$$

$$= \frac{11m}{T} (2m(2L) + 2m(1L) + 7m(-1L))$$

$$= -\frac{11}{T} L = -0.45 \text{ cm} \quad \left(\text{or } -\frac{11}{5} \text{ cm} \right)$$



$$\Delta p_x = m v_{fx} - m v_{ix}$$

$$= m (0 - v_i \cos 35^\circ)$$

$$= 0.30 \text{ kg} (0 - 15 \text{ m/s} \cos 35^\circ)$$

$$= -3.69 \text{ kg} \cdot \text{m/s}$$

$$\Delta p_y = m v_{fy} - m v_{iy}$$

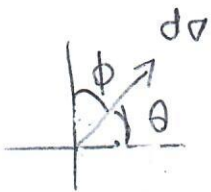
$$= m (-20 \text{ m/s} - (-15 \text{ m/s} \sin 35^\circ))$$

$$= -3.40 \text{ kg} \cdot \text{m/s}$$

$$|\Delta p| = \sqrt{\Delta p_x^2 + \Delta p_y^2} = 5.0 \text{ kg} \cdot \text{m/s}$$

$$\theta = \tan^{-1} \left(\frac{\Delta p_y}{\Delta p_x} \right) = 42.7^\circ$$

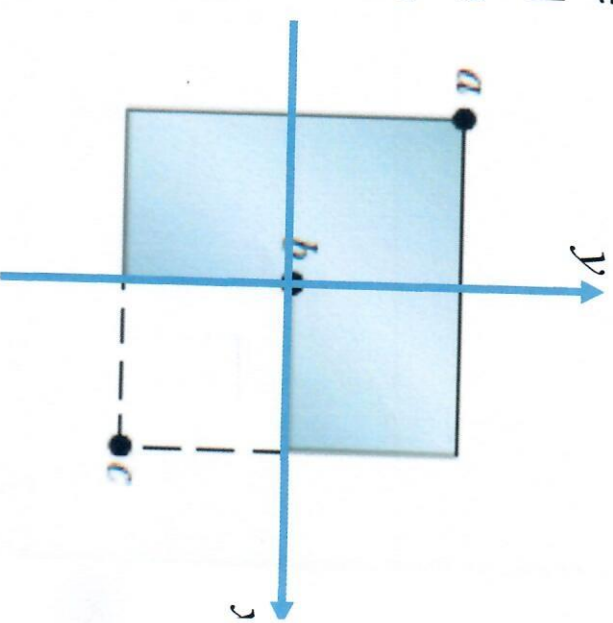
below - x axis



$$\phi = 47.3^\circ$$

1. The figure shows a uniform metal plate that had been square before 25% of it was cut out. Find position of the centre of mass of the metal plate in the figure if the side lengths are a . Use the co-ordinate system shown and give clear reasons for any assumptions you may make.

Die figuur dui 'n uniforme metaal plaat wat oorspronklik vierkantig was voordat 25% daarvan afgesny was. Vind die posisie van die massamiddelpunt van die metaal plaat in die figuur indien die plaat se sy-lengtes het van a . Gebruik die koördinaatstelsel aangedui en gee duidelike verklarings vir enige aannames wat jy maak.

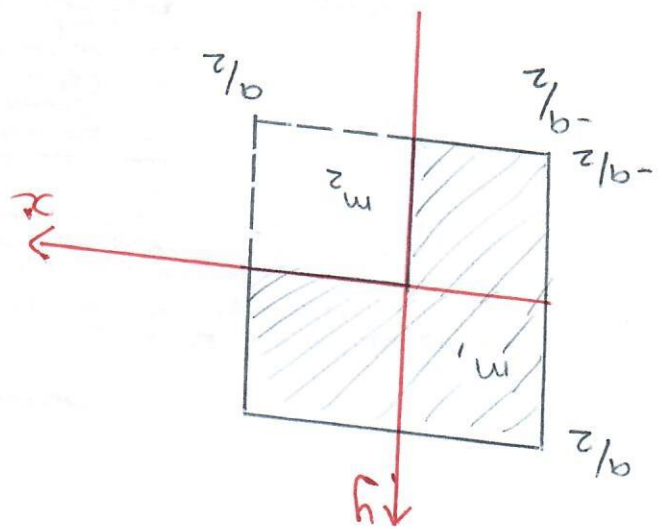


(7)

2. A 70 kg man rides on a 30 kg cart moving at a velocity of 2.5 m/s. He jumps off with zero horizontal velocity relative to the ground. What is the resulting change in the cart's velocity, including sign?

'n 70 kg man ry op 'n 30 kg wa wat teen 'n snelheid van 2.5 m/s beweeg. Jy spring van die wa af met geen horisontale snelheid ten opsigte van die grond. Wat is die verandering in die wa se snelheid, insluitend die teken?

(3)



The original square plate had a com at its centre $(0,0)$
 The cut out has a com at $(\frac{7}{4}a, -\frac{1}{4}a)$

The cut out + the remaining par form the whole and together must have com $(0,0)$

Mass of original square: M
 Mass of cut out: $\frac{M}{4}$
 Mass of remaining piece: $\frac{3M}{4}$

(m_1)
 (m_2)

$$x_{com} = \frac{1}{M} (x_1 m_1 + x_2 m_2) = \frac{1}{M} \left(\frac{3M}{4} x_1 + \frac{M}{4} x_2 \right)$$

$$0 = \frac{3}{4} x_1 + \frac{1}{4} x_2$$

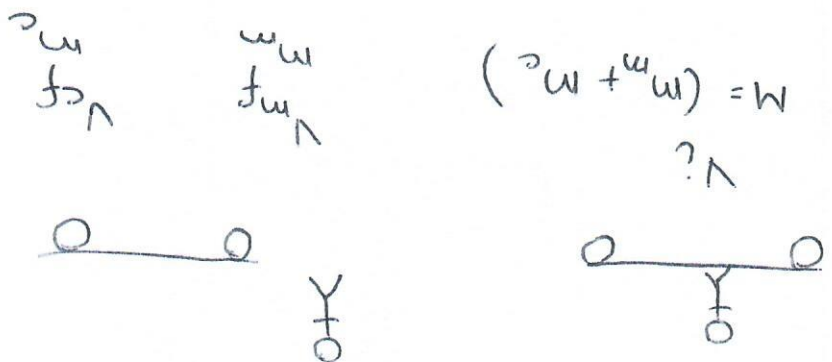
$$x_1 = -\frac{1}{3} x_2 = -0.333 x_2$$

$$y_{com} = \frac{1}{M} (m_1 y_1 + m_2 y_2) = \frac{1}{M} \left(\frac{3M}{4} y_1 + \frac{M}{4} y_2 \right)$$

$$0 = \frac{3}{4} y_1 + \left(-\frac{1}{4} y_2 \right)$$

$$y_1 = \frac{1}{3} y_2 = 0.333 y_2$$

2.



$$M = (m_m + m_c)$$

$V?$

$$\Delta p = 0$$

$$\therefore p_i = p_f$$

$$(m_m + m_c) V_i = m_m V_{mf} + m_c V_{cf}$$

$$V_{cf} = \frac{m_c}{(m_m + m_c)} V_i$$

$$= \frac{30 \text{ kg}}{(70 \text{ kg} + 30 \text{ kg}) 2.5 \text{ m/s}}$$

$$= 8.33 \text{ m/s} = \underline{\underline{8.3 \text{ m/s}}}$$